



Shilpa

**Inclusive Mobile Learning App For Visually,
Hearing, Physically And Cognitively
Impaired Students (Grade 3-5)**

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10/09/2025

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Hearing Impairment Support



BSc (Hons) in Information Technology
Specializing in Information Technology



10/09/2025

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Hearing Impairment Support



Problems & Solutions



Problems

- Most existing systems use ASL/BSL.
- Existing Sinhala apps provide only basic correct/incorrect classification without detailed scoring.
- No engagement features → no games, scoring, or interactive feedback, so students miss motivation (e.g., sign imitation games).
- No Sinhala-first platform offers curriculum syllabus aligned gesture scoring and interactive sign language-based learning for primary education.



Solutions

- Develop a localized platform using Sinhala Sign Language (SLSL), aligned with the Grade 3–5 Sri Lankan curriculum.
- Provide detailed gesture scoring with real-time feedback to improve accuracy, confidence, and learning performance.
- Add sign imitation lesson-based games, quizzes, and badges to increase motivation and provide interactive feedback.
- Enable progress tracking and feedback loops so teachers/parents can support students.



Hearing Impairment Support



Comparison of Research Gaps in Existing System

Feature Research Paper	SriLankan Sign Language(SLSL) Support-Sinhala	Gesture Scoring	Sign Imitation Lesson Based Games	Games	Quizzes
<u>Gesture based Sign Language Recognition system using Mediapipe</u>	(ASL/BSL not SLSL) ❌	✅	❌	❌	❌
<u>Hastha: Online Learning Platform for Hearing Impaired</u>	✅	✅	❌ (text/matching-type games)	✅	✅
<u>Smart System to Support Hearing Impaired Students in Tamil</u>	(Tamil SL) ❌	✅	❌	❌	❌
<u>Visual Kids: Interactive Learning Application for Hearing-Impaired Primary School Kids in Sri Lanka</u>	✅	✅	❌ (typing & text-based learning games)	✅	✅
<u>E-Learning Platform for Hearing Impaired Students</u>	(ASL) ❌	✅	❌	❌	✅
<u>Our Proposed System</u>	✅	✅	✅	✅	✅



Hearing Impairment Support



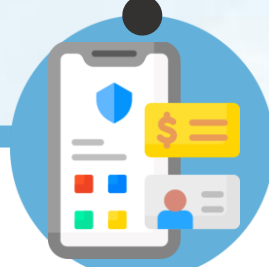
Application of Knowledge & Core Technologies

Application of Knowledge & Core Technologies



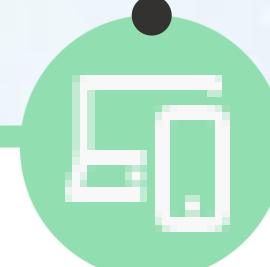
AI/Machine Learning

- Sign imitation pose estimation (hand + body landmarks)-> MediaPipe/BlazePose
- GRU / CNN / DTW models for gesture scoring



HCI & Mobile Accessibility

- Inclusive design for hearing-impaired children
- Simple interaction: visuals, camera input
- Real-time gesture scoring with instant feedback



Technologies & Tools

- Flutter (frontend: lessons, games, quizzes)
- Node.js (backend: storage, progress, quizzes)
- MongoDB (user progress, scores, learning records)
- Cloud storage for videos and reports



Evaluation / Success Metrics

- Accuracy $\geq 85\%$ gesture scoring (validated with certified interpreters)
- Usability testing with Grade 3–5 deaf students
- Engagement: quiz/game completion rate



Data & Ethics

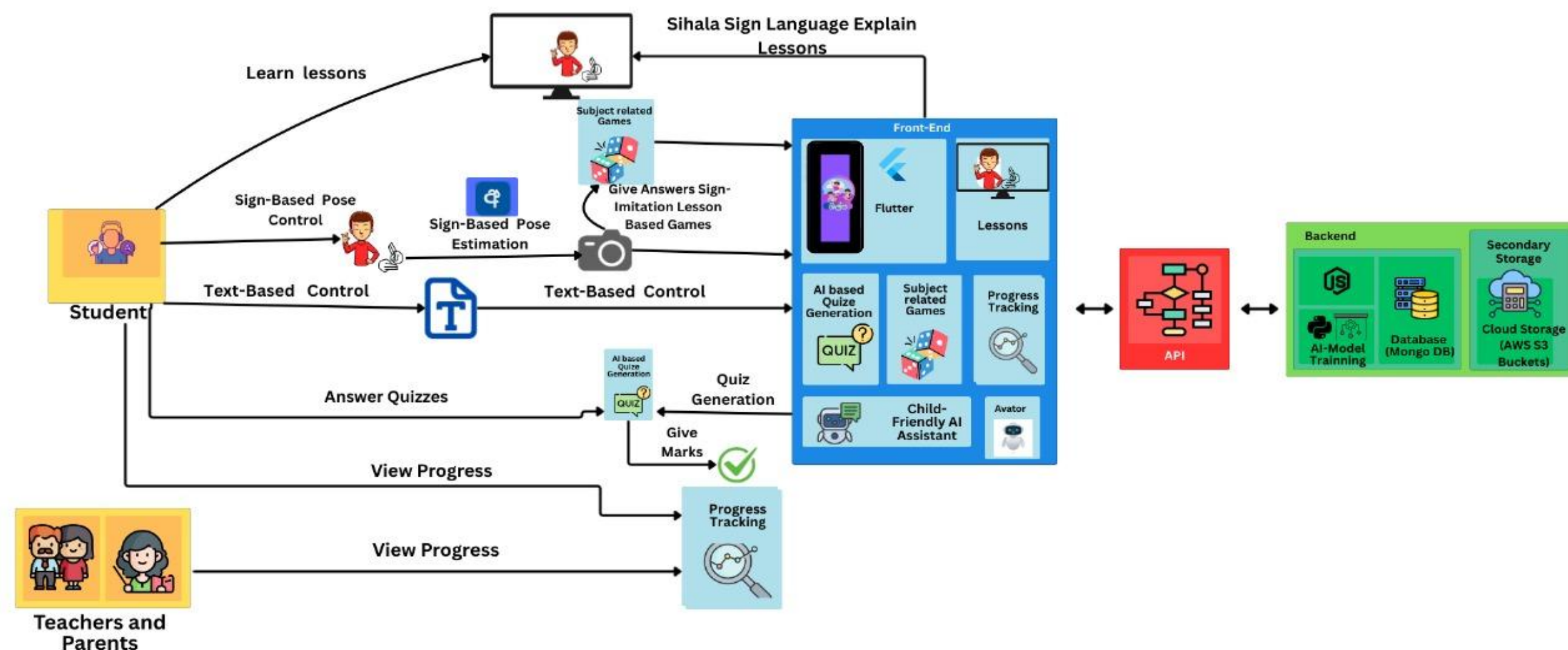
- Sinhala Sign Language syllabus dataset (Tangalle Deaf School + NIE)
- Parental consent & ethical clearance
- Secure storage & anonymized child data



Hearing Impairment Support



System Overview



Real-World Usage

- For Grade 3–5 deaf students (Sri Lanka curriculum)
- Use in classrooms & special schools
- Home learning with parent support
- Lesson-based sign imitation games
- Progress tracking



Visual Impairment Support

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Visual Impairment Support



Problems

- limited to reading text aloud, no curriculum alignment.
- No use of 3D spatial audio for real environment simulation in education.
- Lack of AI-assisted Braille content generation for tactile learning.



Solutions

- AI-powered mobile platform with voice assistant
- Braille-ready docs
- 3D spatial audio storytelling
- vibration feedback



Visual Impairment Support

Comparison of Research Gaps in Existing System

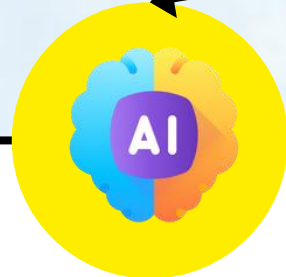


Research Paper	Voice-guided navigation, lessons & quizzes	3D spatial audio learning environments	AI assistant for guided learning	Braille document generation	Vibration feedback for actions
<u>A Study of Educational Pathways of Children with Visual Impairment (Sri Lanka)</u>	✗	✗	✗	✗	✗
<u>An Online Educational Portal for Supporting Open Access to Teaching and Learning of People with Disabilities</u>	✗	✗	✓	✗	✗
<u>Arabic Self-Learning Braille Application</u>	✗	✗	✓	✗	✗
<u>Image-to-Audio Conversion to Aid Visually Impaired (CNN)</u>	✗	✗	✗	✗	✗
<u>Enhancing Accessibility Through Machine Learning: A Review</u>	✗	✗	✓	✗	✗
Our Proposed System	✓	✓	✓	✓	✓

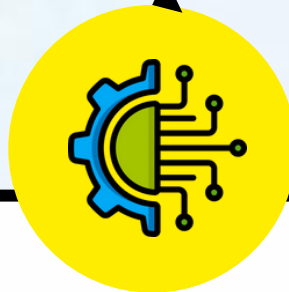


Visual Impairment Support

Application of Knowledge & Technologies



- **Artificial Intelligence (AI)**
TTS, STT, AI voice assistant.
- **Human-Computer Interaction (HCI)**
multi-sensory accessibility design.
- **Cloud & Databases**
Firebase for storage and user profiles.
- **Mobile App Development**
Flutter cross-platform app.



Technology

- Hugging Face TTS/STT models
- Flutter + Dart
- FastAPI backend with Firebase integration
- OpenAL/Unity for 3D spatial sound effects



Evaluation / Success Metrics

- User testing with blind school students (Tangalle).
- Measure:
 1. Accuracy of voice interaction ($\geq 90\%$).
 2. Usability score from VI learners & teachers.
 3. Engagement level with 3D audio lessons.



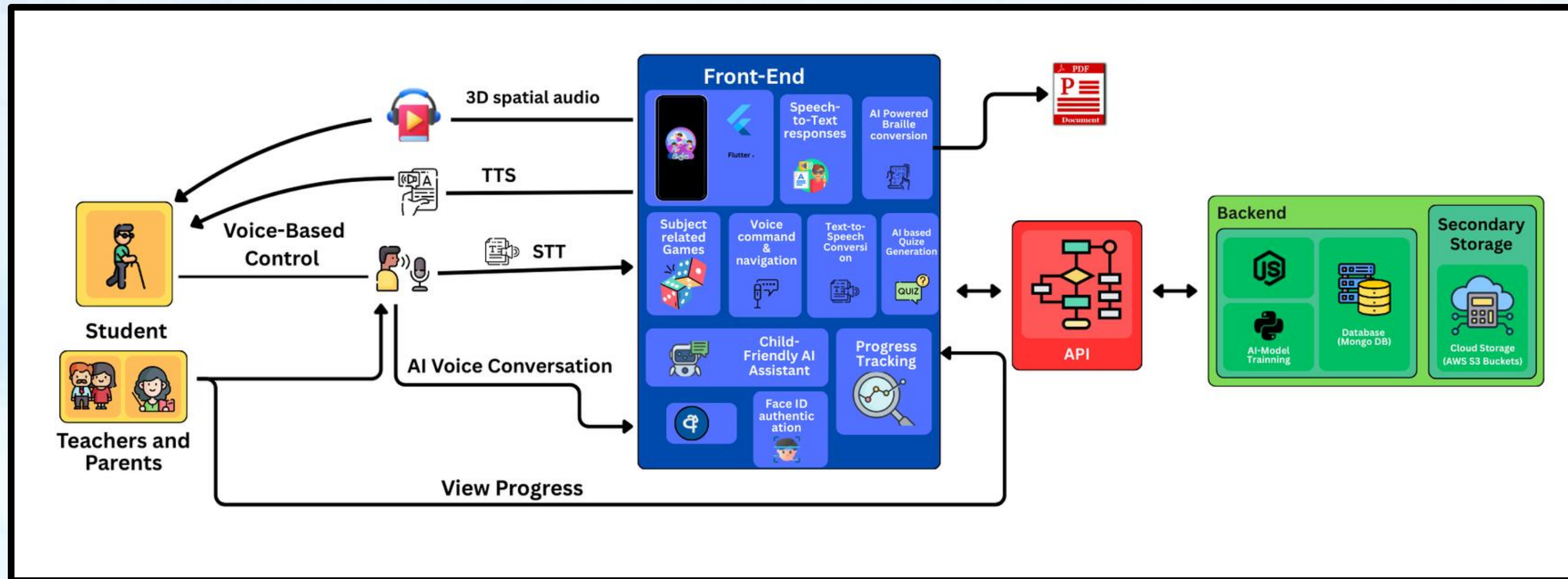
Data & Ethics

- Sinhala Grade 3–5 curriculum, open datasets for TTS/STT
- Required for testing with special needs students.
- Parents & teachers included in evaluation for consent and monitoring.





Visual Impairment Support System Overview



Real-World Usage

- Visually impaired child - learns lessons, answers quizzes by voice.
- Teacher/parent - track student progress.
- Can be used in schools and at home - provides independent learning for VI children.





Physical Impairment Support

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Physical Impairment Support

Problems and Limitations



Problems

- Not providing Safety Features.
- Lack of child friendly environment with simple UIs.
- Lack of multiple options to control (Especially for Physically Disabled People)



Solutions

- Providing multiple handling options
- Customizable Buttons and Fonts
- Providing Accidental Touch protection options and Emergency Stop command feature



Physical Impairment Support

Comparison of Research Gaps in Existing System

Feature Research	Dwell Time Activation	Voice Activation	Gesture / Gaze Track	Multi Modaled	Child Safety UI	Localization
<u>A Mobile App to Optimize Social Participation</u>	✗	✗	✗	✗	✗	✗
<u>Longitudinal eye-gaze performance</u>	✗	✗	✓	✗	✗	✗
<u>Eye-Gaze Control Technology as Early Intervention</u>	✗	✗	✓	✗	✗	✗
<u>photography accessibility (motor impairments)</u>	✗	✗	✗	✗	✗	✗
<u>Smart Wheelchair Robot and Mobile Application</u>	✗	✓	✗	✗	Partial	✓
Our Proposed System	✓	✓	✓	✓	✓	✓



Physical Impairment Support

Application of Knowledge & Technologies



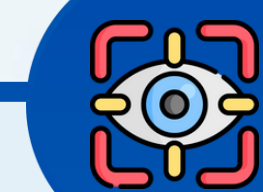
Human Computer Interaction

- **Flutter** - Cross-Platform Framework



Voice Activation

- Hugging Face (wav2vec2)
- Google Cloud Speech-to-Text API



Gaze / Gesture Tracking

- TensorFlow Lite + MediaPipe – real-time gaze/gesture detection.



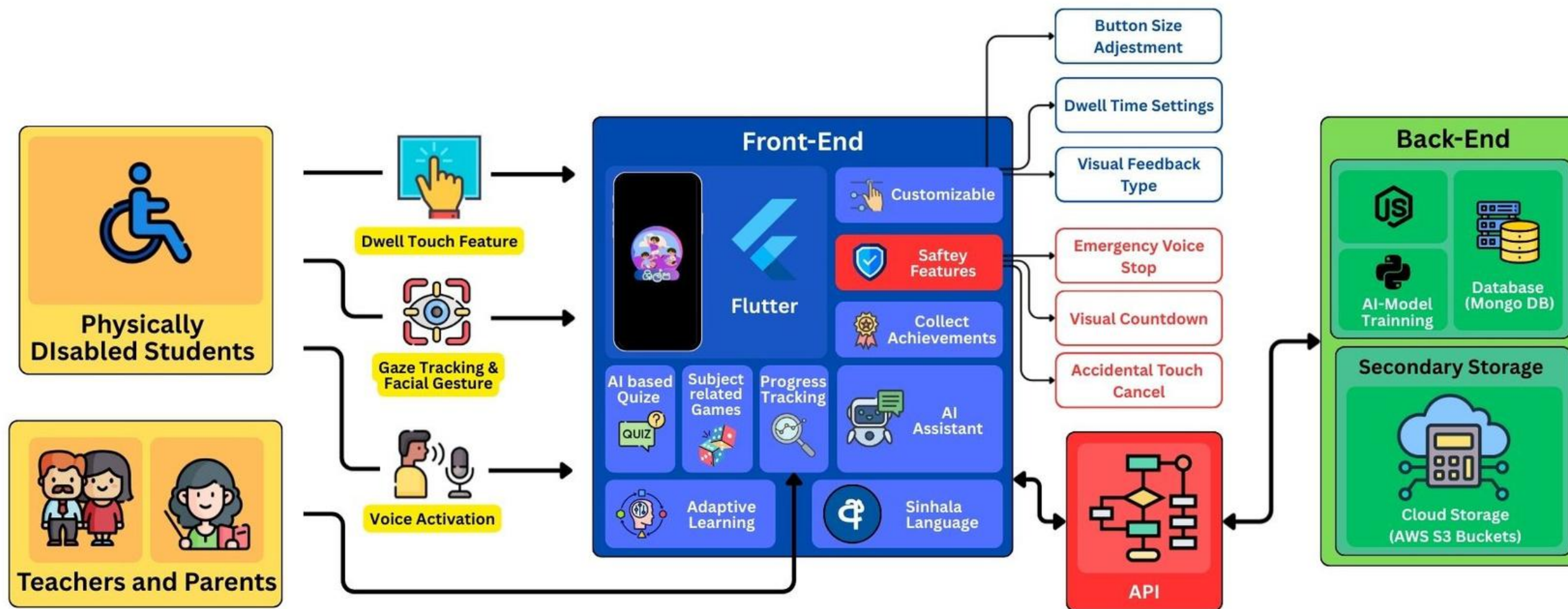
Dwell Time Activation

- Flutter Gesture APIs + Timer Classes → create custom dwell-time controllers.



Physical Impairment Support System Overview

Real-World Usage



- A Grade 3-5 student with limited hand mobility uses **gaze tracking + dwell time** to navigate lessons.
- Emergency stop voice command ("**Stop**") prevents accidental selections.
- A parent of a student **customizes buttons** to be comfortable with the UI



Cognitive & Learning Difficulties Support

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Cognitive & Learning Difficulties Support



Problems

- Sri Lankan students with cognitive disabilities struggle with memory, processing speed, and attention.
- Teaching methods are standardized and do not adapt to varied IQ levels.
- No Sinhala-first adaptive platform that integrates IQ-based personalization, gamification, and emotional guidance.



Solutions

- Animated avatars to provide real-time motivation, encouragement and feedback
- IQ-based adaptive learning paths to match below-average, average, and above-average learners.
- IQ-based adaptive learning paths to match below-average, average and above-average learners.
- Gamification elements (badges, rewards, progress tracking) to increase engagement.



Cognitive & Learning Difficulties Support

Comparison of Research Gaps in Existing System



Features Research	Local Sri Lankan focus	IQ-specific adaptation	Animated Emotional Guidance	Gamified Micro- Activities	Mobile-First + Offline Support
<u>Maggio 2021</u> (Telerehabilitation for LD)	✗	✗	✗	✗	Partial
<u>Looney 2024 (Cognitive Training in Schools)</u>	✗	✗	✗	✗	✗
<u>Guedes ≈2023 (Scaffolding for Inclusive Co-design)</u>	✗	✗	✗	✗	Partial
<u>Paudel 2024 (AT for Dyslexia, AR/VR, NLP)</u>	✗	✗	✗	✓	✓
<u>Hu 2024 (AI-based Psychometric System)</u>	✗	✓	✗	✗	✗
Our Proposed System	✓	✓	✓	✓	✓





Cognitive & Learning Difficulties Support

Application of Knowledge & Core Technologies



AI/Machine Learning

- Adaptive algorithms for IQ-specific learning paths
- Reinforcement learning for activity adjustment.
- TensorFlow Lite – on-device adaptive ML for real-time personalization.



HCI & Mobile Accessibility

- Simplified UI/UX designed for children with cognitive disabilities.
- Universal Design for Learning (UDL) principles applied.
- Compliance with WCAG 2.1 accessibility standards.
- Large buttons, clear visuals, and multimodal delivery (text, audio, visuals).



Tools & Technology

- Flutter SDK – cross-platform app with accessibility widgets.
- Node.js with AWS – backend for authentication, cloud storage, analytics.
- Figma – prototyping accessible, child-friendly user interfaces.



Evaluation & Success Metrics

- Pilot studies with teachers, caregivers, and students in Sri Lankan primary schools.
- Metrics:
 - Higher engagement time vs. non-adaptive lessons.
 - Task completion rates above 80% for adaptive activities.
 - Usability evaluation with System Usability Scale (SUS) target >80.



Data & Ethics

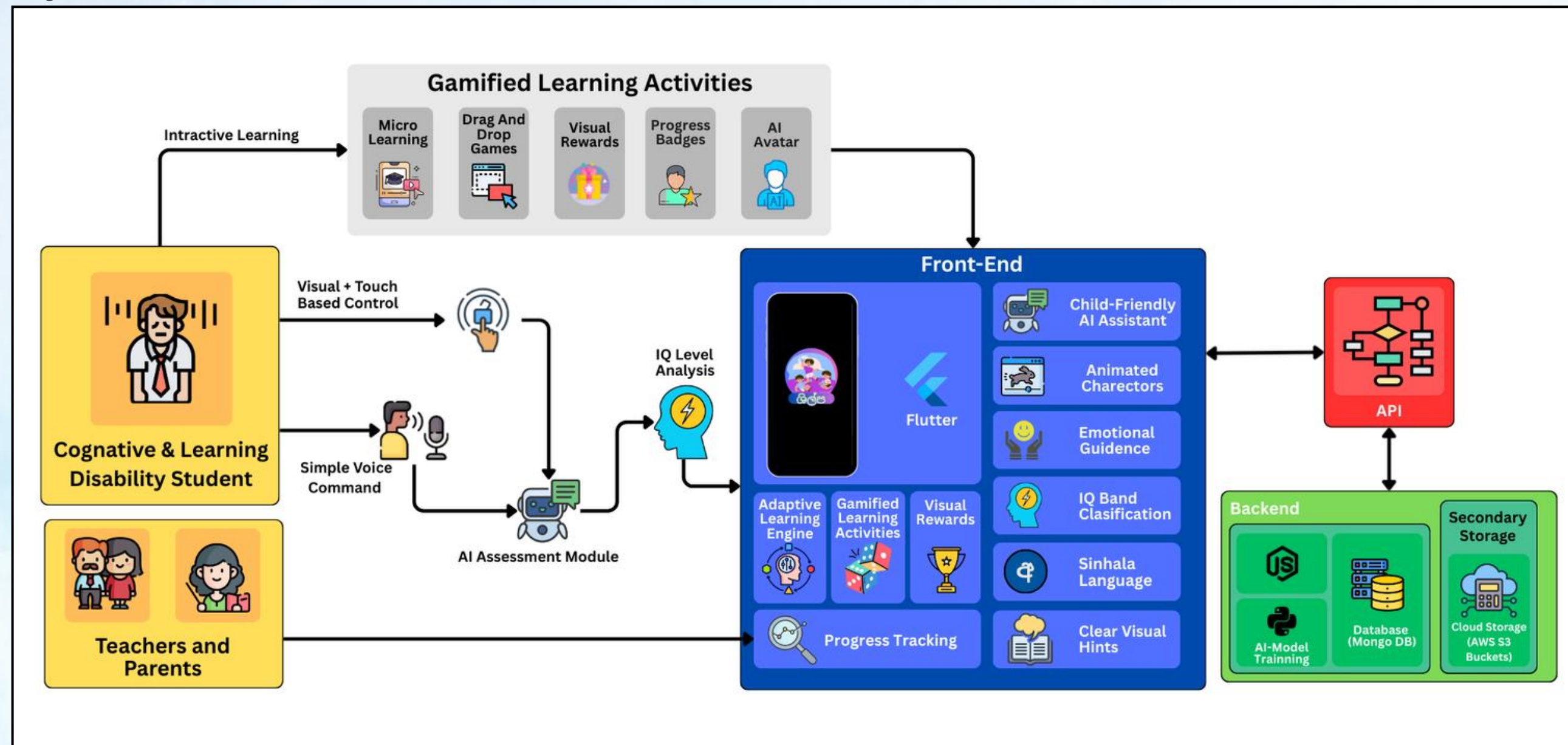
- Primary data from student observations, teacher interviews, and classroom trials.
- Consent obtained from guardians and schools before data collection.
- Anonymization of student profiles to ensure privacy.
- Bias mitigation in IQ-based adaptation to ensure fairness.





Cognitive & Learning Difficulties Support

System Overview



Real-World Usage

- A student logs into the app and begins a micro-activity adapted to their IQ level.
- If the student struggles, the animated avatar provides encouragement and hints in Sinhala.
- Upon completion, the student earns rewards that reinforce motivation.
- Progress data is automatically shared with teachers and parents through real-time dashboards.



Shilpa -Inclusive Mobile Learning App For Visually, Hearing, Physically And Cognitively Impaired Students (Grade 3-5)

Ability of Commercialization

- Sinhala curriculum-aligned mobile platform supporting Visual, Hearing, Physical, and Cognitive impairments.
- High adoption potential in Sri Lankan schools, and special education centers.

Investment / Cost & Recovery

- Labor Costs: App development (Flutter, Firebase, AI/ML), research, dataset collection, and external supervisor consultations.
- Material Costs: cloud hosting, video production.

Target Market

390,000+ Sri Lankan students with disabilities

Value Proposition

- Localized
- Progress Tracking

Property Rights

- Sinhala Sign Language dataset
- AI models for adaptive learning
- Game-based learning content.

